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Biochemical test chip

Abstract

The present disclosure provides a biochemical test chip, including an electrode unit and a protective layer. The protective layer is electrically connected to the electrode unit. The protective layer is configured to oxidize the electrode unit after the electrode unit receives an electron or reduce the electrode unit after the electrode unit loses an electron. There is a potential difference (E.sub.cell.sup.0) between the protecting layer and the electrode unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefits of Taiwan Patent Application No. 109141392, filed on Nov. 25, 2020, which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present application relates to a biochemical test chip for use in medical measurement, particularly to a biochemical test chip including a protective layer.

BACKGROUND

(3) In-vitro medical measurement plays a vital role in today's medical industry; by qualitatively and quantitatively measuring biological fluids changes, it provides index information for rapid diagnosis and treatment of diseases. The use of biochemical test chips has become a standard technique for medical or biochemical testing.

(4) Conventional biochemical test chips have at least two electrodes. After loading the specimen into the biochemical test chip's reaction zone, the specimen's electrochemical properties can be measured using said two electrodes. For most electrochemical test strips, a reactive layer reacts with a specimen, and the electrochemical properties of the specimen are analyzed by electrochemical means. However, the composition of the reactive layer is extremely sensitive to environmental factors such as temperature, humidity, and oxygen level. For example, the reactive layer's conductive medium is susceptible to oxidation with oxygen in the air. This oxidation reaction can easily cause inaccurate measurement of electrochemical test strips, leading to the short service life of conventional electrochemical test strips.

(5) To address the effect of environmental factors on electrochemical test strips, electrochemical test strips are often stored in containers with desiccant, protected from light, and relatively sealed to avoid damage to electrochemical test strips. However, when the user opens the container to retrieve the electrochemical test strips, the container's internal environment will change when the container is opened, and hence, the other electrochemical test strips stored in the container may be affected. For example, when the electrodes in an electrochemical test strip are exposed to air, water vapor, or other environments, the electrode surface is susceptible to oxidation or reduction reactions that reduce its electron transfer capability and/or increase its impedance. This can still cause inaccurate measurements and reduce the lifetime of the electrochemical test strips.

(6) The “prior art” discussion above merely provides a technology background without acknowledging that the “prior art” discussed above reveal the subject matter of this disclosure and do not constitute prior art at this time and that any of the “prior art” discussion above should not be regarded as any part of the present application.

SUMMARY OF THE INVENTION

(7) The present disclosure provides a biochemical test chip, including an electrode unit and a protective layer. The protective layer is electrically connected to the electrode unit. The protective layer is configured to oxidize the electrode unit after the electrode unit receives an electron or reduce the electrode unit after the electrode unit loses an electron. There is a potential difference ($E_{\text{sub.cell.sup.0}}$) between the protective layer and the electrode unit.

(8) In some embodiments, the potential difference ($E_{\text{sub.cell.sup.0}}$) is greater than 0.

(9) In some embodiments, the protective layer is an anode, and the electrode unit is a cathode.

(10) In some embodiments, the protective layer is a cathode, and the electrode unit is an anode.

(11) In some embodiments, the biochemical test chip further includes a first insulating septum, located on the electrode unit, wherein the first insulating septum has a first opening and a second opening, wherein the first opening at least partially exposes the electrode unit, and the second opening at least partially exposes the protective layer.

(12) In some embodiments, the biochemical test chip further includes a second insulating septum, located on the first insulating septum, wherein the second insulating septum has a third opening, and the third opening at least partially exposes the protective layer.

(13) In some embodiments, the electrode unit includes a branch, wherein the branch is configured to provide a conductive platform to the protective layer.

(14) In some embodiments, the electrode unit and the protective layer are substantially located on the same level.

(15) In some embodiments, at least a portion of the electrode unit is made of an active material and is in contact with air.

- (16) In some embodiments, the protective layer is configured to protect the portion made of the active material or the whole electrode unit.
- (17) The present disclosure provides a biochemical test chip, including a reactive layer, an electrode unit, and a protective layer. The reactive layer is electrically connected to the electrode unit. The protective layer is electrically connected to the electrode unit, and the protective layer is electrically connected to the reactive layer via the electrode unit. The protective layer is configured to oxidize the reactive layer after the reactive layer receives an electron or reduce the reactive layer after the reactive layer loses an electron, wherein there is a potential difference (E.sub.cell.sup.0) between the protective layer and the reactive layer.
- (18) In some embodiments, the potential difference (E.sub.cell.sup.0) is greater than 0.
- (19) In some embodiments, the protective layer is an anode, and the reactive layer is a cathode.
- (20) In some embodiments, the protective layer is a cathode, and the reactive layer is an anode.
- (21) In some embodiments, an oxidation or reduction reaction current level of the reactive layer is greater than an oxidation or reduction reaction current level of the protective layer.
- (22) In some embodiments, an oxidation or reduction reaction current level of the reactive layer is greater than or equal to 10-fold an oxidation or reduction reaction current level of the protective layer.
- (23) The present disclosure's biochemical test chip is disposed with a protective layer, wherein the protective layer can be used to maintain the stability of the biochemical test chip's reactive layer, so as to protect the biochemical test chip, delay or avoid the unwanted spoilage of the biochemical test chip with the environment, thereby prolonging the shelf-life of the biochemical test chip. Besides, the biochemical test chip can be disposed with a plurality of protective layers to protect different components of the biochemical test chip. For example, the protective layer can further protect the working electrode or the counter electrode and the like so as to decrease the measurement error. Moreover, the biochemical test chip can further include a detachable protective layer so as to accelerate the process for the biochemical test chip to return to the default state.
- (24) The foregoing outlines the technical features and advantages of the present disclosure so that those skilled in the art may better understand the following detailed description of the present application. Other technical features and advantages that constitute the subject matter of the present disclosure are described below. Those skilled in the art should appreciate that they may readily use the concepts and specific embodiments provided below as a basis for designing or modifying other structures and processes for carrying out the same purposes and/or achieving the same advantages of the present disclosure. Those skilled in the art should also realize that such equivalent constructions still fall within the present disclosure's spirit and scope as defined in the appended claims.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description and claims when read with the accompanying figures. It is noted that elements with the same reference numbers are the same elements.
- (2) FIG. 1 is a schematic exploded view illustrating a biochemical test chip according to some embodiments of the present disclosure.
- (3) FIG. 2 is a partial top view illustrating a biochemical test chip according to some embodiments of the present disclosure
- (4) FIG. 3 is a schematic cross-sectional view taken from the line A-A' in FIG. 2.
- (5) FIG. 4 is a schematic cross-sectional view taken from the line A-A' in FIG. 2.
- (6) FIG. 5 is a schematic exploded view illustrating a biochemical test chip according to some embodiments of the present disclosure.
- (7) FIG. 6 is a schematic exploded view illustrating a biochemical test chip according to some embodiments of the present disclosure.
- (8) FIG. 7 is a schematic exploded view illustrating a biochemical test chip according to some embodiments of the present disclosure.
- (9) FIG. 8 is a partial top view illustrating a biochemical test chip according to some embodiments of the present disclosure
- (10) FIG. 9 is a partial top view illustrating a biochemical test chip according to some embodiments of the

present disclosure

(11) FIG. 10A and FIG. 10B provide test results from samples with five different blood glucose concentrations to illustrate the difference between the present disclosure and conventional techniques.

DETAILED DESCRIPTION

(12) Detailed description of the present disclosure is discussed in detail below. However, it should be understood that the embodiments provide many inventive concepts that can be applied in a variety of specific contexts. The specific embodiments discussed are illustrative of the specific ways they can be made and used and do not limit the present disclosure's scope.

(13) The same reference numeral is configured to represent the same elements/components in the various drawings and illustrative embodiments. Reference will now be made in detail to the illustrative embodiments shown in the drawings. Whenever possible, the same reference numeral is used in the drawings and the specification to represent the same or similar parts. In the drawings, the shape and thickness may be exaggerated for clarity and convenience. The description will be directed specifically to the elements forming part of, or more directly cooperating with, the device disclosed hereunder. As could be appreciated, elements not explicitly shown or described may take various forms. The reference to "some embodiments" or "embodiment" throughout this specification implies that the particular features, structures, or characteristics described in conjunction with the embodiment are included in at least one of the embodiments. Therefore, the phrase "in some embodiments" or "in an embodiment" appearing in various places throughout this specification does not necessarily refer to the same embodiment. Besides, the specific features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

(14) In the drawings, the same reference numeral is configured to indicate the same or similar elements in the various views, and illustrative embodiments of the present application are shown and described. The drawings are not necessarily drawn to scale, and in some cases, the drawings have been exaggerated and/or simplified and are configured for illustrative purposes only. Many possible applications and variations of the present application will be understood by those of ordinary skill in the art in view of the following illustrative embodiments of the present disclosure.

(15) Unless otherwise defined, all terms used herein, including technical and scientific terms, have the same meanings as those commonly understood by a person of ordinary skill in the art in the field of the disclosed embodiments. It should be understood, for example, that terms defined in common dictionaries should be construed to have meanings consistent with their meanings in the relevant field and context of this disclosure and should not be construed or understood to have meanings that are too formal unless expressly defined herein.

(16) Besides, the following embodiments are provided to illustrate the core value of this disclosure but are not intended to limit the scope of protection of this disclosure. For clarity and ease of understanding, the same or similar functions or elements among this disclosure's different embodiments are not repeated or shown in the drawings. Besides, different elements or technical features from different embodiments may be combined or substituted to create further embodiments that are still covered by this disclosure, provided they do not conflict with each other.

(17) The present disclosure is directed to an electrochemical system that sacrifices itself to protect a specific object, particularly to a biochemical test chip of the electrochemical system that sacrifices itself to protect a specific object, in particular. Furthermore, the present disclosure is directed to a biochemical test chip, including a protective layer, wherein the protective layer can sacrifice itself to maintain the stability of one or more components in the biochemical test chip. In some embodiments, the protective layer is configured to protect the object in the biochemical test chip that is protected by the protective layer, such as less-stable components in the biochemical test chip or those susceptible to oxidation or reduction. The subject that is protected by the protective layer can be, for example, the electrode unit or reactive layer in the biochemical test chip; however, the present disclosure is not limited thereto. In detail, the present disclosure slows or avoids one or more components in the biochemical test chip to suffer from spoilage other than medical measurements (e.g., unwanted spoilage upon being exposed to the environment) by sacrificing the protective layer (such as by allowing the protective layer to undergo the oxidation or reduction reaction). Depending on the chemical properties of the object to be protected, the protective layer sacrifices itself to undergo the corresponding scarification (e.g., the oxidation or reduction reaction) to facilitate said component(s) to return to the default state or to provide or receive electrons before the component is spoiled, thereby attaining the technical effect of protecting the components.

(18) For example, when it is expected to keep the object to be protected stably in the reduced state, materials

capable of providing electrons are chosen to form the protective layer. In other words, when the object to be protected is spoiled because it loses electrons, if the protective layer is made from materials capable of providing or losing electrons, then the protective layer can lose electrons first or compensate for the electrons that the object to be protected loses before the object to be protected loses electrons. In this way, it is feasible to keep the object to be protected stably in the reduced state.

(19) On the other hand, when it is expected to keep the object to be protected stably in the oxidized state, materials capable of receiving electrons are chosen to form the protective layer. In other words, when the object to be protected is spoiled because it receives electrons, if the protective layer is made from materials capable of capturing or accommodating electrons, then the protective layer can obtain electrons first before the object to be protected receives electrons. In this way, it is feasible to keep the object to be protected stably in the oxidized state.

(20) Generally, substances with a higher standard reduction potential tend to receive electrons, whereas substances with a lower standard reduction potential tend to lose electrons. According to the Gibbs Free Energy relationship, i.e., $\Delta G_{sup.0} = -nFE_{cell, sup.0}$, where $\Delta G_{sup.0}$ is the change in the free energy, n is the mole number of electrons, and F is the charge per mole. The equation for the potential difference ($E_{cell, sup.0}$) is $E_{cell, sup.0} = E_{cathode} - E_{anode}$, where $E_{cathode}$ is the standard reduction potential of a cathode (cathode electrode), and E_{anode} is the standard reduction potential of an anode (anode electrode). When the Gibbs free energy $\Delta G_{sup.0} < 0$, the reaction is spontaneous. From the above, it can be seen that when two oxidizable/reduceable substances with a potential difference ($E_{cell, sup.0}$) are in the same reaction tank, the one with higher standard reduction potential will tend to undergo reduction reaction, and the other one will tend to undergo oxidation reaction. For example, when the standard reduction potential of the anode is smaller than that of the cathode, the anode will spontaneously transfer electrons to the cathode, and the cathode will remain in the reduced state because it continues to receive electrons, thus avoiding the influence of environmental oxidants (e.g., oxygen, water vapor, etc.).

(21) Therefore, by disposing the protective layer and the object to be protected in the same reaction tank, and allowing the two to have a potential difference ($E_{cell, sup.0}$), wherein the potential difference ($E_{cell, sup.0}$) is greater than 0, it is feasible to allow the protective layer and the object to be protected to generate an electron flow in a specific direction spontaneously so that the object to be protected can be kept in its original redox state. In this way, the biochemical test chip is protected to slow or avoid the unwanted spoilage of the biochemical test chip with the environment.

(22) Depending on the protective layer's materials and the subject to be protected, the protective layer and the object to be protected can respectively be an anode and a cathode, and the protective layer and the object to be protected can also be a cathode and an anode, respectively. The following paragraphs provide several embodiments of the present disclosure, which are used as examples to illustrate the core value of the present disclosure; however, they are not used to limit the protection scope of the present disclosure.

(23) Reference is made to FIG. 1, FIG. 1 is a schematic exploded view illustrating a biochemical test chip **100** according to some embodiments of the present disclosure. The biochemical test chip **100** can be an electrochemical test chip, which is a device that can be electrically connected to. The biochemical test chip **100** is configured to collect a specimen and carry out an electrochemical reaction therewith so as to detect a target analyte therein. The specimen includes any liquids or soluble solids having therein one target analyte that can be detected using an electrochemical method. For example, the specimen may include blood, tissue fluid, urine, sweat, tears, and other biological samples; however, the present disclosure is not limited thereto. Moreover, the blood can include the whole blood, plasma, serum, etc.; however, the present disclosure is not limited thereto.

(24) Reference is made to FIG. 1; the biochemical test chip **100** includes an insulating substrate **10**, an electrode unit **20**, a first insulating septum **30**, a reactive layer **40**, a second insulating septum **50**, and the protective layer **60**. The insulating substrate **10** includes a substrate that is electrically insulated. In some embodiments, the material of the insulating substrate **10** can include polyvinyl chloride (PVC), glass fiber (FR-4), polyethersulfone (PES), bakelite, polyethylene terephthalate (PET), polycarbonate (PC), polypropylene (PP), polyethylene (PE), polystyrene (PS), polyimide (PI), glass plate, ceramic or any combination of the above-mentioned materials; however, the present disclosure is not limited thereto. The material of the insulating substrate **10** can be adjusted depending on the system or actual needs.

(25) The electrode unit **20** of the biochemical test chip **100** is located on the insulating substrate **10**. The electrode unit **20** is disposed on the insulating substrate **10** and configured to be subjected to the electrochemical measurement. The electrochemical measurement includes analyzing the specimen's

concentration using an electrical reaction, such as potentiometry, conductometry, voltammetry, polarimetry, high-frequency titration, amperometry, Coulombic method, electrolysis, and the like. The electrode unit **20** includes a working electrode **22** and a counter electrode **24**; however, the present disclosure is not limited thereto. The electrode unit **20** can have other electrodes depending on the requirements of the system. The working electrode **22** is the electrode that allows the conductive medium to undergo the electrooxidation reaction or electroreduction reaction on the surface thereof and can be used by the measuring apparatus to determine the concentration. In detail, the electrooxidation reaction or electroreduction reaction is an electrochemical reaction in which the conductive medium undergoes an exchange between electrical and chemical energy on the surface of the working electrode **22**.

(26) The polarity of the working electrode **22** can be an anode or a cathode, depending on the requirement of the measurement reaction. For example, if the conductive medium is oxidized on the working electrode **22**, the working electrode **22** is an anode; if the conductive medium is reduced on the working electrode **22**, the working electrode **22** is a cathode. The counter electrode **24** is an electrode that undergoes the electroreduction reaction or electrooxidation reaction corresponding to the working electrode **22** so that the overall electrochemical system satisfies the principles of charge balance. The potential and polarity of the counter electrode **24** are opposite to the potential and the polarity of the working electrode **22**. Before being in contact with the specimen, the working electrode **22** and the counter electrode **24** are insulated from each other. After the working electrode **22** and the counter electrode **24** are in contact with the specimen, they form an electrical loop with the measuring apparatus. In some embodiments, the working electrode **22** and the counter electrode **24** can include a carbon electrode, silver electrode, platinum electrode, etc.; however, the present disclosure is not limited thereto.

(27) The materials of the working electrode **22** and the counter electrode **24** can vary depending on the system's requirement.

(28) The first insulating septum **30** is disposed on the insulating substrate **10** and located on the electrode unit **20**. The first insulating septum **30** can have an opening **32**, wherein the opening **32** at least partially exposes the electrode unit **20**. In some embodiments, the opening **32** is located at the front side **30F** of the first insulating septum **30** and exposes a portion of the electrode unit **20**. The opening **32** is configured to define a reaction zone **34** in the biochemical test chip **100**; the reaction zone **34** is configured to accommodate the specimen. The electrode unit **20** is exposed from the portion of the opening **32** and can undergo the electrochemical reaction with the specimen. The size or shape of the opening **32** can be adjusted according to the desired area of the electrode unit **20** and the desired volume of the specimen. In some embodiments, the backside **30B** of the first insulating septum **30** exposes a portion of the electrode unit **20** to form a connecting zone **10C**. The electrode unit **20** exposed from the connecting zone **10C** can be electrically connected to a measuring apparatus (not shown in the drawings). The measuring apparatus and the biochemical test chip **100** are electrically connected to provide the energy required for the electrochemical measurement and analyze the reaction signal. In some embodiments, the material of the first insulating septum **30** includes a PVC insulation tape, PET insulation tape, heat drying insulation paint, or ultraviolet (UV) curable insulation paint; however, the present disclosure is not limited thereto.

(29) FIG. 2 is a partial top view illustrating the biochemical test chip **100** according to some embodiments of the present disclosure. Reference is made to FIG. 2 and FIG. 1 simultaneously; the biochemical test chip **100** further includes a reactive layer **40**. The reactive layer **40** is located in the opening **32** of the first insulating septum **30**. The reactive layer **40** is electrically connected to the electrode unit **20**. In some embodiments, the area of the reactive layer **40** is smaller than the size of the opening **32**. The reactive layer **40** at least partially covers the electrode unit **20** exposed from the opening **32**. In some embodiments, the reactive layer **40** covers both the working electrode **22** and the counter electrode **24**. In some embodiments, the reactive layer **40** only covers the working electrode **22**. The reactive layer **40** is configured to undergo a chemical reaction with the specimen. In other embodiments, the biochemical test chip **100** does not include the reactive layer **40**.

(30) In some embodiments, the reactive layer **40** includes an enzyme and a conductive medium. For example, the enzyme includes a fixed or non-fixed enzyme, such as redox enzymes, antigens, antibodies, microbial cells, animal and plant cells, and biologically identifiable components of animal and plant tissues. The conductive medium is configured to receive electrons generated after the reaction between the enzyme and a blood specimen and transmit the electrons to the measuring apparatus via the electrode unit **20**. In some embodiments, the conductive medium can include potassium hexacyanoferrate(III), potassium hexacyanoferrate(II) trihydrate, ruthenium complex, ferrocene, sodium dithionite, nicotinamide adenine

dinucleotide (NAD⁺), nicotinamide adenine dinucleotide phosphate (NADP⁺), thiamin pyrophosphate (TPP), coenzyme A (HSCoA), flavin adenine dinucleotide (FAD) or a combination thereof; however, the present disclosure is not limited thereto. In some embodiments, the reactive layer **40** can be further supplemented with a phosphate buffer and protectants, such as a protein, dextrin, glucan, amino acid, etc.; however, the present disclosure is not limited thereto.

(31) Reference is made again to FIG. **1**; the second insulating septum **50** is located on the first insulating septum **30**. In some embodiments, the second insulating septum **50** at least partially covers the opening **32** of the first insulating septum **30** so that the opening **32** forms a capillary structure. In some embodiments, the terminus of the second insulating septum **50** is disposed with a vent **502** corresponding to the opening **32**. The vent **52** can be of any shape; for example, the vent **52** can be circular, oval, rectangular, rhombus, etc. In some embodiments, the second insulating septum **50** also exposes the connecting zone **10C** of the electrode unit **20**. The second insulating septum **50** can be of any shape or size.

(32) Reference is made to FIG. **2** and FIG. **1** simultaneously; the biochemical test chip **100** further includes the protective layer **60**. For example, water vapor or oxygen and the like in the environment may exist between the insulating substrate **10** and the first insulating septum **30**. Therefore, the reactive layer **40** exposed to the environment and the electrode unit **20** disposed between the insulating substrate **10** and the first insulating septum **30** may get spoiled upon being oxidized with the water vapor or oxygen in the air.

(33) In the present embodiment, the protective layer **60** can be configured to protect the reactive layer **40** in the biochemical test chip **100** to slow or avoid the unwanted spoilage of the biochemical test chip **100** in the environment.

(34) In some embodiments, as shown in FIG. **1**, the first insulating septum **30** can have an opening **34**, and the second insulating septum **50** can have an opening **54**, wherein the opening **34** and the opening **54** at least partially expose the protective layer **60**. The protective layer **60** and the reactive layer **40** can be exposed to the same environment; however, the present disclosure is not limited thereto. For example, the protective layer **60** can be disposed between the insulating substrate **10** and the first insulating septum **30** and is exposed to the same environment as the reactive layer **40** via the opening **34** and the opening **54**. In other embodiments, the protective layer **60** and the reactive layer **40** are exposed to different environments. For example, the protective layer **60** can be disposed between the insulating substrate **10** and the first insulating septum **30**, and the first insulating septum **30** and the second insulating septum **50** do not have the opening **34** and opening **54**.

(35) The protective layer **60** is disposed at a specific region of the electrode unit **20**. For example, the protective layer **60** is disposed on the working electrode **22**; however, the present disclosure is not limited thereto. As shown in FIG. **1**, the protective layer **60** is disposed on the electrode unit **20** and is electrically connected to the electrode unit **20**. The protective layer **60** and the reactive layer **40** are electrically coupled via the electrode unit **20**. In the present embodiment, the protective layer **60** is electrically connected to the reactive layer **40** via the working electrode **22**; however, the present disclosure is not limited thereto. In other embodiments, the protective layer **60** can be electrically connected to the reactive layer **40** via the counter electrode **24**. The position of the protective layer **60** is not limited to those described above; in some embodiments, the protective layer **60** can be disposed on the second insulating septum **50** and electrically connected to the electrode unit **20** via wires. In other embodiments, the protective layer **60** can be disposed between the first insulating septum **30** and the second insulating septum **50** and electrically connected to the electrode unit **20** via wires.

(36) The protective layer **60** can take the forms of solid, liquid, or gas. For example, solids can include pure metals, alloys, metal compounds (halides, oxides, mixed-valence compounds, organometallic complexes), organic redox agents, etc. Liquids can include aqueous solutions, organic solutions, supercritical fluids, liquid elements (e.g., bromine, mercury), etc. Gases can include gaseous elements (e.g., oxygen, ozone), gaseous compounds (e.g., ammonium, nitrogen monoxides) etc.

(37) In the present embodiment, the protective layer **60** is configured to protect the reactive layer **40** in the biochemical test chip **100**. The protective layer **60** and the reactive layer **40** can have different materials or compositions. For example, the protective layer **60** and the reactive layer **40** can have different standard reduction potentials, and hence, there is a potential difference ($E_{\text{sub.cell.sup.0}}$) between the protective layer **60** and the reactive layer **40**. The potential difference ($E_{\text{sub.cell.sup.0}}$) between the protective layer **60** and the reactive layer **40** is greater than 0. The protective layer **60** and the reactive layer **40** are in the same reaction environment. In some embodiments, the protective layer **60** and the reactive layer **40** are in contact with air simultaneously; however, the present disclosure is not limited thereto. The protective layer **60** and

the reactive layer **40** electrically connected to the electrode unit **20** simultaneously can also be considered the two being in the same reaction environment. Because there is a potential difference ($E_{\text{sub.cell.sup.0}}$) between the protective layer **60** and the reactive layer **40**, and the potential difference ($E_{\text{sub.cell.sup.0}}$) is greater than 0, an electron flow in a specific direction is generated spontaneously so that the object to be protected (the reactive layer **40**) can be kept in its original redox state. In this way, the biochemical test chip **100** is protected to slow or avoid the unwanted spoilage of the biochemical test chip **100** with the environment.

(38) Depending on the materials of the protective layer **60** and the reactive layer **40**, the protective layer **60** and the reactive layer **40** can respectively be an anode and a cathode, and the protective layer **60** and the reactive layer **40** can also be a cathode and an anode, respectively. In the following paragraphs, FIG. **3** and FIG. **4** are respectively used to discuss the examples where the protective layer **60** is the cathode and the reactive layer **40** is the anode, and the protective layer **60** is the anode and the reactive layer **40** is the cathode. In some embodiments, the area of the protective layer **60** is greater than the area of the reactive layer **40**. In some embodiments, the area of the protective layer **60** is substantially the same as the area of the reactive layer **40**. The area and the thickness of the protective layer **60** and the reactive layer **40** can be adjusted depending on the system requirements.

(39) Reference is made to FIG. **3**; FIG. **3** is a schematic cross-sectional view taken from the line A-A' in FIG. **2**. In the present embodiment, the conductive medium of the reactive layer **40** can be ferricyanide (hexacyanoferrate(III)); however, ferricyanide tends to be reduced into ferrocyanide (hexacyanoferrate(II)) upon light irradiation, wherein the reaction equation can be expressed as

$\text{Fe}^{\text{sup.III}}(\text{CN})_{\text{sub.6}}^{\text{sup.3-}} \xrightarrow{\text{hv}} \text{Fe}^{\text{sup.II}}(\text{CN})_{\text{sub.6}}^{\text{sup.4-}}$. When ferricyanide is reduced into ferrocyanide conductive medium, this spoilage results in the increases of the background current when the biochemical test chip **100** performs the concentration measurement, thereby causing measurement errors.

(40) As shown in FIG. **2** and FIG. **3**, the biochemical test chip **100** of the present disclosure is disposed with the protective layer **60**, wherein the protective layer **60** is electrically connected to the reactive layer **40** via the electrode unit **20**. Thus, when the biochemical test chip **100** is exposed to the environment and before the specimen is loaded, the reactive layer **40**, the electrode unit **20**, the protective layer **60**, and the air form a chemical reaction tank. In the present embodiment, the material of the protective layer **60** can be silver oxide (silver peroxide). Silver oxide can undergo the reduction reaction and give silver, wherein the reaction equation can be expressed as $\text{Ag}_2\text{O} + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{Ag} + \text{H}_2\text{O}$. The standard reduction potential of silver oxide/silver is about 1.77 V, and the standard reduction potential of ferricyanide/ferrocyanide is about 0.36 V. Therefore, in the present embodiment, the protective layer **60** is the cathode, whereas the reactive layer **40** is the anode. The potential difference ($E_{\text{sub.cell.sup.0}}$) between the protective layer **60** and the reactive layer **40** is 1.41 V. Since the potential difference between the two is greater than 0, the change in free energy is smaller than 0, and thus, ferrocyanide in the reactive layer **40** undergoes spontaneous oxidation reaction into ferricyanide, wherein the reaction equation can be expressed as $2\text{Fe}^{\text{sup.II}}(\text{CN})_{\text{sub.6}}^{\text{sup.4-}} + \text{Ag}_2\text{O} + 2\text{H}^+ \rightarrow 2\text{Fe}^{\text{sup.III}}(\text{CN})_{\text{sub.6}}^{\text{sup.3-}} + 2\text{Ag} + \text{H}_2\text{O}$. In this case, the half-reaction taking place on the reactive layer **40** is

$\text{Fe}^{\text{sup.II}}(\text{CN})_{\text{sub.6}}^{\text{sup.4-}} \rightarrow \text{Fe}^{\text{sup.III}}(\text{CN})_{\text{sub.6}}^{\text{sup.3-}} + \text{e}^-$.

(41) In view of the above, ferrocyanide in the reactive layer **40** is oxidized into ferricyanide because of the reduction reaction of silver oxide in the protective layer **60**. When the reduction reaction of the protective layer **60** takes place, the reactive layer **40** undergoes the oxidation reaction, thereby mitigating the reduction reaction due to the light irradiation. Therefore, by disposing the protective layer **60** in the biochemical test chip **100**, it is feasible to effectively avoid the conductive medium in the reactive layer **40** from being spoiled before measuring the specimen. The composition materials of the conductive medium of the reactive layer **40** and the protective layer **60** are not limited to those discussed above. In some embodiments, the composition materials of the conductive medium of the reactive layer **40** and the protective layer **60** are selected so that the potential difference ($E_{\text{sub.cell.sup.0}}$) between the two is greater than 0.

(42) In the present embodiment, a method for protecting the biochemical test chip **100** is provided. In detail, the present method for protecting the biochemical test chip **100** includes providing the protective layer **60** electrically connected to the electrode unit **20**, wherein the protective layer **60** is electrically connected to the reactive layer **40** via the electrode unit **20**. The protective layer **60** is configured to oxidize the reactive layer **40** after the reactive layer **40** receives electrons. As discussed above, ferricyanide in the reactive layer **40** tends to receive electrons and thus is reduced into ferrocyanide upon light irradiation. Hence, one may choose a material with a standard reduction potential greater than the standard reduction potential of

ferricyanide/ferrocyanide as the protective layer 60; for example, one may choose silver oxide as the protective layer 60. Therefore, after the reactive layer 40 receives electrons, the protective layer 60 allows the reactive layer 40 to undergo the oxidation reaction so that the reactive layer 40 returns to the default state. In the present embodiment, the standard reduction potential of the reactive layer 40 is smaller than the standard reduction potential of the protective layer 60; however, the present disclosure is not limited thereto.

(43) Reference is made to FIG. 4; FIG. 4 is a schematic cross-sectional view taken from the line A-A' in FIG. 2. In the present embodiment, the standard reduction potential of the reactive layer 40 is greater than the standard reduction potential of the protective layer 60. In the present embodiment, the conductive medium of the reactive layer 40 can be ferrocyanide; however, ferrocyanide, when being exposed to the air, tends to be oxidized by oxygen into ferricyanide, wherein the reaction equation can be expressed as $\text{Fe}^{\text{sup.II}}(\text{CN})_{\text{sub.6}}^{\text{sup.4-}} + \text{O}_{\text{sub.2}} + 2\text{H}^{\text{sup.+}} \rightarrow \text{Fe}^{\text{sup.III}}(\text{CN})_{\text{sub.6}}^{\text{sup.3-}} + \text{H}_{\text{sub.2}}\text{O}_{\text{sub.2}}$. When ferrocyanide is oxidized into ferricyanide after it is in contact with oxygen, this spoilage results in the change of the conductive medium's concentration in the biochemical test chip 100, thereby affecting the background current and causing measurement errors.

(44) As shown in FIG. 2 and FIG. 4, the biochemical test chip 100 of the present disclosure is disposed with the protective layer 60, wherein the protective layer 60 is electrically connected to the reactive layer 40 via the electrode unit 20. Thus, when the biochemical test chip 100 is exposed to the environment and before the specimen is loaded, the reactive layer 40, the electrode unit 20, the protective layer 60, and air form a chemical reaction tank. In the present embodiment, the material of the protective layer 60 can be iron. When being exposed to the air, iron tends to undergo oxidation reaction with water vapor, wherein the half-reaction equation can be expressed as $\text{Fe} + 2\text{OH}^{\text{sup.-}} \rightarrow \text{Fe}(\text{OH})_{\text{sub.2}} + 2\text{e}^{\text{sup.-}}$. The standard reduction potential of ferricyanide/ferrocyanide is about 0.36 V, whereas the standard reduction potential of iron(II) hydroxide/iron is about -0.89 V. Thus, in the present embodiment, the reactive layer 40 is the cathode, and the protective layer 60 is the anode. When the biochemical test chip 100 is exposed to the environment having water vapor and oxygen, the potential difference ($E_{\text{sub.cell}}^{\text{sup.0}}$) between the protective layer 60 and the reactive layer 40 is 1.25 V. Since the potential difference between the two is greater than 0, the change in free energy is smaller than 0; and hence, ferricyanide in the reactive layer 40 undergoes the spontaneous reduction reaction into ferrocyanide, wherein the reaction equation can be expressed as $2\text{Fe}^{\text{sup.III}}(\text{CN})_{\text{sub.6}}^{\text{sup.3-}} + \text{Fe} + 2\text{OH}^{\text{sup.-}} \rightarrow 2\text{Fe}^{\text{sup.II}}(\text{CN})_{\text{sub.6}}^{\text{sup.4-}} + \text{Fe}(\text{OH})_{\text{sub.2}}$. In this case, the half-reaction on the reactive layer 40 can be expressed as $\text{Fe}^{\text{sup.III}}(\text{CN})_{\text{sub.6}}^{\text{sup.3-}} + \text{e}^{\text{sup.-}} \rightarrow \text{Fe}^{\text{sup.II}}(\text{CN})_{\text{sub.6}}^{\text{sup.4-}}$.

(45) Hence, ferricyanide in the reactive layer 40 is reduced into ferrocyanide because of iron's oxidation reaction in the protective layer 60. When the oxidation reaction of the protective layer 60 takes place, the reactive layer 40 undergoes the reduction reaction, thereby slowing the oxidation reaction due to oxygen in the air. Besides, the reactive layer 40 is protected by the oxidation of the water vapor in the air to iron so as to keep the desired stability of the conductive medium of the reactive layer 40. Therefore, by disposing the protective layer 60 in the biochemical test chip 100, it is feasible to effectively avoid the conductive medium in the reactive layer 40 from being spoiled before measuring the specimen. The composition materials of the conductive medium of the reactive layer 40 and the protective layer 60 are not limited to those discussed above.

(46) In the present embodiment, a method for protecting the biochemical test chip 100 is provided. In detail, the present method for protecting the biochemical test chip 100 includes providing the protective layer 60 electrically connected to the electrode unit 20, wherein the protective layer 60 is electrically connected to the reactive layer 40 via the electrode unit 20. The protective layer 60 is configured to reduce the reactive layer 40 after the reactive layer 40 loses electrons. As discussed above, ferrocyanide in the reactive layer 40 tends to be oxidized into ferricyanide by oxygen when being exposed to the air. Hence, one may choose a material with a standard reduction potential smaller than the standard reduction potential of ferricyanide/ferrocyanide as the protective layer 60; for example, one may choose iron as the protective layer 60. Therefore, after the reactive layer 40 loses electrons, the protective layer 60 allows the reactive layer 40 to undergo the reduction reaction so that the reactive layer 40 returns to the default state. In the present embodiment, the standard reduction potential of the reactive layer 40 is greater than the standard reduction potential of the protective layer 60; however, the present disclosure is not limited thereto.

(47) The foregoing are examples of the protective layer 60 and the reactive layer 40, and the present disclosure is not limited thereto. By choosing appropriate materials as the protective layer 60, it is feasible that the potential difference ($E_{\text{sub.cell}}^{\text{sup.0}}$) between the protective layer 60 and the reactive layer 40 is

greater than 0. In this way, the protective layer **60** and the reactive layer **40** are under spontaneous reactions so as to protect the biochemical test chip **100** and prolong the shelf-life of the biochemical test chip **100**.

(48) The material of the protective layer **60** can be determined based on the material of the reactive layer **40**. For example, when it is desired to stabilize the conductive medium of the reactive layer **40** in the oxidized state, materials suitable for use as a cathode are chosen as the material of the protective layer **60**, i.e., materials capable of receiving electrons can be used. In other words, when the reactive layer **40** spoils upon receiving electrons, materials capable of capturing or accommodating electrons are chosen as the material of the protective layer **60**; hence, the protective layer **60** receives electrons first before the reactive layer **40** receives electrons. In this way, the reactive layer **40** is stabilized in the oxidized state. Generally, substances with a higher standard reduction potential tend to receive electrons, whereas substances with a lower standard reduction potential tend to lose electrons. When it is expected to keep the reactive layer **40** stably in the oxidized state, materials with a standard reduction potential greater than that of the reactive layer **40** are chosen to form the protective layer **60**.

(49) When it is expected to keep the reactive layer **40** stably in the reduced state, materials suitable for use as an anode are chosen as the material of the protective layer **60**; i.e., materials capable of providing electrons can be used. In other words, when the reactive layer **40** is spoiled because it loses electrons, if the protective layer **60** is made from materials capable of providing or losing electrons, then the protective layer **60** can lose electrons first or compensate for the electrons that the reactive layer **40** loses before the reactive layer **40** loses electrons. In this way, it is feasible to keep the reactive layer **40** stably in a reduced state. As discussed above, when it is expected to keep the reactive layer **40** stably in the reduced state, materials with a standard reduction potential lower than that of the reactive layer **40** are chosen to form the protective layer **60**.

(50) Generally, in addition to the reactive layer **40**, components in the biochemical test chip **100** that are susceptible to redox reaction further include the working electrode **22** and the counter electrode **24**. In some embodiments, the biochemical test chip **100** can be disposed with an additional protective layer to protect the working electrode **22** or the counter electrode **24**. In some embodiments, the protective layer **60** can protect the working electrode **22** and the reactive layer **40** simultaneously. In some embodiments, the protective layer **60** must be in physical contact with the object to be protected to provide the protection. The subject to be protected can include the reactive layer **40**, the working electrode **22**, the counter electrode **24**, or other units in the biochemical test chip **100**. In other embodiments, the protective layer **60** can protect the protection by merely being electrically coupled to the object to be protected. In some embodiments, when the protective layer **60** is in contact with an object to be protected that is made of compounding materials, one should avoid the counterreaction on the parts other than the object to be protected. Hence, the protective layer **60** is designed to be in electrical contact only with the object to be protected.

(51) It should be noted that in order to prevent the protective layer **60** from interfering with the biochemical test chip **100** during the electrochemical measurement, the protective layer **60** according to the present disclosure is disconnected during the concentration measurement reaction. In some embodiments, as shown in FIG. 2, the protective layer **60** is connected in series with the reactive layer **40** via the working electrode **22** and the branch **22A** thereof. In some embodiments, the protective layer **60** can be connected to the multiple branches of the working electrode **22** so as to form a parallel connection with the reactive layer **40**. In some embodiments, the biochemical test chip **100** may be connected to a circuitry switch that disconnects the protective layer **60** through the circuitry switch while the biochemical test chip **100** is under the electrochemical measurement. When the biochemical test chip **100** is electrically connected to the measuring instrument, the protective layer **60** is electrically separated from the electrode unit **20**. Besides, the biochemical test chip **100** is electrochemically measured by providing a reaction potential through an external meter to drive the concentration measurement reaction. The order of magnitude of the chemical reactions in the reaction layer **40** will be much higher than the oxidation or reduction chemical reactions between the protective layer **60** and the reaction layer **40**. Thus, the protective layer **60** will not affect the concentration measurement reaction of the biochemical test chip **100**. In some embodiments, the oxidation or reduction reaction current level of the reaction layer **40** is greater than or equal to 10 times (inclusive) the oxidation or reduction reaction current level of the protective layer **60**. In some embodiments, the oxidation or reduction reaction current level of the reaction layer **40** is greater than or equal to 50 times (inclusive) the oxidation or reduction reaction current level of the protective layer **60**. In some embodiments, the oxidation or reduction reaction current level of the reaction layer **40** is greater than or equal to 100 times (inclusive) the oxidation or reduction reaction current level of the protective layer **60**. For example, to ensure that the oxidation or reduction reaction of the protective layer **60** does not interfere with the measurement reaction

(i.e., the reaction of the reaction layer **40**), if the minimum oxidation or reduction reaction current level of the reaction layer **40** (i.e., the lowest concentration oxidation or reduction reaction current level on the system specification) is 1 microampere (μA), the oxidation or reduction reaction current level of the protective layer **60** should be less than $0.01\ \mu\text{A}$. Thus, the oxidation or reduction reaction current level of the reaction layer **40** is preferably greater than or equal to more than 100 times (inclusive) the oxidation or reduction reaction current level of the protective layer **60**. In some embodiments, the role of the protective layer **60** relative to the reaction layer **40** may also change so that the protective layer **60** does not affect the results of the concentration measurement reaction.

(52) In some embodiments, the branch **22A** of the working electrode **22** is provided to provide the protective layer **60** a conductive platform. In some embodiments, the working electrode **22** and the branch **22A** thereof may have the same or different materials. In some embodiments, the branch **22A** of the working electrode **22** may include carbon, such as a carbon layer. In other embodiments, the working electrode **22** may not have a branch **22A**, and the protective layer **60** may be provided directly on the working electrode **22**.

(53) The present disclosure is not limited to the foregoing embodiments and can comprise other different embodiments. For simplification purposes and to facilitate the comparison among embodiments of the present disclosure, in the following embodiments, each of the completely the same elements is labeled with completely the same reference numeral. To further facilitate the comparison among the differences between these embodiments, only the differences among different embodiments are discussed, whereas the completely the same features are not discussed for the sake of brevity.

(54) In some embodiments, the protective layer **60** can be configured to protect the electrode unit **20**. Reference is made to FIG. 5; FIG. 5 is a schematic exploded view illustrating a biochemical test chip **200** according to some embodiments of the present disclosure. As shown in FIG. 5, the difference between the biochemical test chip **200** and the biochemical test chip **100** is that the electrode unit **20** and the protective layer **60** locate at substantially the same level. In some embodiments, only a portion of the electrode unit **20** is disposed on the protective layer **60**. For example, the electrode unit **20** includes a branch, wherein the branch is disposed on the protective layer **60**. The branch can be the branch **22A** of the working electrode **22**, the branch **24A** of the counter electrode **24** (shown in FIG. 8), or the branch of a reference electrode. In some embodiments, the branch can be configured to provide the protective layer **60** a conductive platform.

(55) In the present embodiment, the working electrode **22** of the electrode unit **20** is disposed on the protective layer **60**, and the counter electrode **22** of the electrode unit **20** and the protective layer **60** are disposed on the same level; however, the present disclosure is not limited thereto. In some embodiments, only the branch **22A** of the working electrode **22** in the electrode unit **20** is disposed on the protective layer **60**. For example, the working electrode **22** and the protective layer **60** locate at substantially the same level; only the branch **22A** of the working electrode **22** is disposed on the protective layer **60**. In the present embodiment, the first insulating septum **30** and the second insulating septum **50** do not have the opening **34** and the opening **54**; however, the present disclosure is not limited thereto. In other embodiments, openings may be disposed on the first insulating septum **30** and the second insulating septum **50** depending on the system requirements. For example, the first insulating septum **30** can have the opening **34**, and the second insulating septum **50** can have the opening **54**, wherein the opening **34** and the opening **54** at least partially expose the branch **22A** of the working electrode **22**.

(56) In some embodiments, the electrode unit **20** completely covers the protective layer **60**, and the reactive layer **40** is disposed on the electrode unit **20**. In detail, the working electrode **22** of the electrode unit **20** completely covers the protective layer **60**. In some embodiments, the branch **22A** of the working electrode **22** can completely cover the protective layer **60**. In some embodiments, the area of the branch **22A** of the working electrode **22** is greater than the area of the protective layer **60**. In some embodiments, the thickness of the working electrode **22** is greater than the thickness of the protective layer **60**. In some embodiments, the thickness of the working electrode **22** is substantially the same as the thickness of the protective layer **60**.

(57) As shown in FIG. 5, the biochemical test chip **200** is disposed with the protective layer **60**, and the protective layer **60** is electrically connected to the electrode unit **20**. In some embodiments, the protective layer **60** is configured to protect the electrode unit **20**. In some embodiments, the protective layer **60** is configured to protect part of or the whole electrode unit **20**. In some embodiments, the protective layer **60** is configured to protect the portion of the electrode unit **20** that is made of the active material or the whole electrode unit. The active material can include a conductive material, such as silver or other suitable metal materials. In the present embodiment, the protective layer **60** is configured to protect the working electrode **22** of the electrode unit **20**. For example, the material of the working electrode **22** can include silver;

however, silver tends to react with oxygen and water vapor in the air and get oxidized into silver oxide, wherein the reaction equation can be expressed as $4\text{Ag} + \text{O}_2 \rightarrow 2\text{Ag}_2\text{O}$. When silver is oxidized into silver oxide after being exposed to the air, it will result in the toxification on the surface of the working electrode **22**; thereby reducing the working electrode **22**'s conductivity and its capability to receive electrons; such spoilage will result in measurement errors when the biochemical test chip **200** is subject to the concentration measurement.

(58) The biochemical test chip **200** according to the present disclosure is disposed with the protective layer **60**, wherein the protective layer **60** is electrically connected to the electrode unit **20**. Thus, when the biochemical test chip **200** is exposed to the environment and before the specimen is loaded, the electrode unit **20**, the protective layer **60**, and air form a chemical reaction tank. In the present embodiment, the material of the protective layer **60** can include ferrocene, wherein the ferric iron located at the center of ferrocene is susceptible to valence transition and undergoes the redox reaction, wherein the reaction equation can be expressed as $\text{Fe}^{\text{II}}(\text{C}_5\text{H}_5)_2 \rightarrow \text{Fe}^{\text{III}}(\text{C}_5\text{H}_5)_2 + e^-$, and the standard reduction potential thereof is about 0.16 V. The standard reduction potential of silver oxide/silver is about 1.17 V. The standard reduction potential of ferrocene is smaller than the standard reduction potential of silver oxide/silver. Thus, in the present embodiment, the protective layer **60** is the anode, whereas the electrode unit **20** is the cathode. The potential difference (E_{cell}) between the protective layer **60** and the electrode unit **20** is about 1.01 V. Since the potential difference between the two is greater than 0, the change in free energy is smaller than 0, and hence, silver oxide in the electrode unit **20** undergoes the spontaneous reduction reaction into silver, wherein the reaction equation can be expressed as $2\text{Fe}^{\text{II}}(\text{C}_5\text{H}_5)_2 + \text{Ag}_2\text{O} + 2\text{H}^+ \rightarrow 2\text{Fe}^{\text{III}}(\text{C}_5\text{H}_5)_2 + 2\text{Ag} + \text{H}_2\text{O}$.

(59) In view of the above, silver oxide in the working electrode **22** is reduced to silver because of ferrocene's oxidation reaction in the protective layer **60**. When the protective layer **60** is oxidized, the working electrode **22** of the electrode unit **20** undergoes the reduction reaction, thereby slowing the oxidation reaction of oxygen and water vapor in the air. Therefore, by disposing the protective layer **60** in the biochemical test chip **200**, one can effectively prevent the electrode unit **20** from spoilage before measuring the specimen. The composition materials of the electrode unit **20** and the protective layer **60** are not limited thereto.

(60) The material of the protective layer **60** is not limited to ferrocene. In some embodiments, the material of the protective layer **60** can include a metallocene or a macrocycle; however, the present disclosure is not limited thereto. For example, the metallocene can include vanadocene, chromocene, manganocene, ferrocene, cobaltocene, nickelocene, rhodocene, etc. The macrocycle can include iron phthalocyanine, cobalt phthalocyanine, nickel phthalocyanine, copper phthalocyanine, etc.

(61) In the present embodiment, a method for protecting the biochemical test chip **200** is provided. In detail, the present method for protecting the biochemical test chip **200** includes providing the protective layer **60** electrically connected to the electrode unit **20**. The protective layer **60** is configured to reduce the electrode unit **20** after the electrode unit **20** loses electrons. As discussed above, silver in the electrode unit **20** tends to be oxidized into silver oxide by oxygen when being exposed to the air. Hence, one may choose a material with a standard reduction potential smaller than the standard reduction potential of silver/silver oxide as the protective layer **60**; for example, one may choose ferrocene as the protective layer **60**. Therefore, after the electrode unit **20** loses electrons, the protective layer **60** allows the electrode unit **20** to undergo the reduction reaction so that the electrode unit **20** returns to the default state. In the present embodiment, the standard reduction potential of the electrode unit **20** is greater than the standard reduction potential of the protective layer **60**; however, the present disclosure is not limited thereto.

(62) In other embodiments, the standard reduction potential of the electrode unit **20** is smaller than the standard reduction potential of the protective layer **60**. For example, the present disclosure further provides a method for protecting the biochemical test chip **200**, including providing the protective layer **60** that is electrically connected to the electrode unit **20**. The protective layer **60** is configured to oxidize the electrode unit **20** after the electrode unit **20** receives electrons. As discussed above, one can choose a material with a standard reduction potential greater than the standard reduction potential of the electrode unit **20** as the protective layer **60**. Therefore, after the electrode unit **20** receives electrons, the protective layer **60** allows the electrode unit **20** to undergo the oxidation reaction so that the electrode unit **20** returns to the default state.

(63) The foregoing is only an example of the protective layer **60** and the electrode unit **20**, and the present disclosure is not limited thereto. By choosing appropriate materials as the protective layer **60**, the potential difference (E_{cell}) between the protective layer **60** and the electrode unit **20** can be greater than 0.

In this way, the protective layer **60** and the electrode unit **20** are in spontaneous reactions, thereby protecting the biochemical test chip **200** and prolonging the shelf-life of the biochemical test chip **200**.

(64) Reference is made to FIG. 6; FIG. 6 is a schematic exploded view illustrating a biochemical test chip **300** according to some embodiments of the present disclosure. As shown in FIG. 6, the difference between the biochemical test chip **300** and the biochemical test chip **100** is that the electrode unit **20** of the biochemical test chip **300** does not include the branch **22A**. In some embodiments, the protective layer **60** can be directly disposed on the working electrode **22** of the electrode unit **20**; however, the present disclosure is not limited thereto. In other embodiments, the protective layer **60** can be directly disposed on the counter electrode **24** of the electrode unit **20**.

(65) Reference is made to FIG. 7; FIG. 7 is a schematic exploded view illustrating a biochemical test chip **400** according to some embodiments of the present disclosure. As shown in FIG. 7, the difference between the biochemical test chip **400** and the biochemical test chip **100** is that the biochemical test chip **400** includes a detachable protective layer **70**, instead of the protective layer **60**. Generally, during the production or manufacturing process of the biochemical test chip **400**, components in the biochemical test chip **400** may be spoiled due to the manufacturing environment. For example, the reactive layer **40** is in the liquid form before being dried, and the electrode unit **20** is subject to a high-temperature electrode manufacturing process; this unavoidable environment of high temperature, high humidity, and light irradiation will result in the spoilage of the biochemical test chip **400**, thereby leading to measurement errors. Thus, in some embodiments, the biochemical test chip **400** further includes the detachable protective layer **70**. When the biochemical test chip **400** is at rest, the detachable protective layer **70** can be used to restore the spoiled unit to the default state. It is worth noting that the detachable protective layer **70** of the present embodiment refers to a protective layer that is not disposed in the biochemical test chip **400**, and the detachable protective layer **70** is replaceable and can be removed before being connected with a measuring apparatus subjected to measurement reaction. In some embodiments, an external force (e.g., applying a voltage) can be applied on the detachable protective layer **70** to accelerate the restoration of the biochemical test chip **400**.

(66) In some embodiments, the detachable protective layer **70** is electrically connected to the electrode unit **20** so as to restore the biochemical test chip **400** to the default state. As shown in FIG. 7, the detachable protective layer **70** is covered on the electrode unit **20** via the insulation tape **72**. After the biochemical test chip **400** returns to the default state, the detachable protective layer **70** can electrically isolate the electrode unit **20**. For example, the detachable protective layer **70** can be removed by stripping off the insulation tape **72**. In some embodiments, the detachable protective layer **70** is electrically connected to the reactive layer **40**. In some embodiments, the detachable protective layer **70** is connected to the counter electrode **24**.

(67) Reference is made to FIG. 8; FIG. 8 is a partial top view illustrating a biochemical test chip **500** according to some embodiments of the present disclosure. As shown in FIG. 8, the difference between the biochemical test chip **500** and the biochemical test chip **100** is that the protective layer **60** of the biochemical test chip **500** is disposed on one side of the counter electrode **24**. In the present embodiment, the protective layer **60** is electrically connected to the counter electrode **24**. The protective layer **60** is configured to protect the counter electrode **24**. In some embodiments, the protective layer **60** can be disposed on the counter electrode **24**. For example, the protective layer **60** can be disposed on the branch **24A** of the counter electrode **24**. In some embodiments, the protective layer **60** can also be disposed under the counter electrode **24**. The relative positions of the protective layer **60** and the electrode unit **20** can be adjusted depending on the system requirements.

(68) Reference is made to FIG. 9; FIG. 9 is a partial top view illustrating a biochemical test chip **600** according to some embodiments of the present disclosure. As shown in FIG. 9, the difference between the biochemical test chip **600** and the biochemical test chip **100** is that the biochemical test chip **600** can include multiple protective layers **60**, **62**, such as the protective layer **60** and the protective layer **62**. The protective layer **60** is electrically connected to the working electrode **22**, whereas the protective layer **62** is electrically connected to the counter electrode **24**. The protective layer **60** is configured to protect the working electrode **22** or the reactive layer **40**, whereas the protective layer **62** is configured to protect the counter electrode **24**. In the present embodiment, the reactive layer **40** is only in contact with the working electrode **22** but is not in contact with the counter electrode **24**. Therefore, the protective layer **60** is electrically connected to the reactive layer **40** via the working electrode **22**.

(69) The biochemical test chip **600** can protect different regions in the biochemical test chip **600** by disposing multiple protective layers. For example, the protective layer **60** is configured to protect the stability of the reactive layer **40**, whereas the protective layer **62** is configured to protect the counter electrode's stability **24**.

In other embodiments, the protective layer 60 is configured to protect the stability of the working electrode 22, whereas the protective layer 62 is configured to protect the stability of the counter electrode 24. The protective layer 60 and the protective layer 62 can have the same or different materials. Moreover, the protective layer 60 and the protective layer 62 can have different areas; however, the present disclosure is not limited thereto. In some embodiments, the area of the protective layer 60 is greater than the area of the protective layer 62. In some embodiments, the protective layer 60 and the protective layer 62 locate at a different level; however, the present disclosure is not limited thereto. For example, the protective layer 60 can locate on the working electrode 22, whereas the protective layer 62 can locate under the counter electrode 24.

(70) FIG. 10A and FIG. 10B provide test results from samples with five different blood glucose concentrations to illustrate the difference between the present disclosure and conventional techniques. In detail, FIG. 10A and FIG. 10B show the signal regression analysis of five replicate experiments of five blood glucose specimens (100 mg/dL, 200 mg/dL, 300 mg/dL, 400 mg/dL, and 500 mg/dL) measured by biochemical test chip with accelerated aging at 50° C. for four weeks and a biochemical analyzer. FIG. 10A shows the signal of the biochemical specimen without the protective layer, and FIG. 10B shows the signal of the biochemical specimen with the protective layer. The solid lines in FIG. 10A and FIG. 10B are standard values.

(71) As shown in FIG. 10A, when the blood glucose concentration is 100 mg/dL, the signal expression of biochemical test chip without the protective layer is relatively high. Moreover, when the blood glucose concentration is greater than 300 mg/dL (inclusive), the signals of the biochemical test chip without the protective layer are all lower than the biochemical analyzer and all exceed the acceptable range of $\pm 10\%$. As shown in FIG. 10B, when the blood glucose concentration is lower than 300 mg/dL (inclusive), the signals obtained by the biochemical test chip according to the present disclosure are relatively concentrated and are similar to the biochemical analyzer results. Moreover, at all the blood glucose, the signals obtained by the biochemical test chip according to the present disclosure are all within the acceptable range.

(72) The foregoing discussion of the present disclosure provides a biochemical test chip disposed with a protective layer, wherein the protective layer can be used to maintain the stability of the biochemical test chip's reactive layer so as to protect the biochemical test chip, delay or avoid the unwanted spoilage of the biochemical test chip with the environment, thereby prolonging the shelf-life of the biochemical test chip. Besides, the biochemical test chip can be disposed with a plurality of protective layers to protect different components of the biochemical test chip. For example, the protective layer can further protect the working electrode or the counter electrode and the like so as to decrease the measurement error. Moreover, the biochemical test chip can further include a detachable protective layer so as to accelerate the process for the biochemical test chip to return to the default state.

(73) Although the disclosure and its advantages have been described in detail, it should be understood that various modifications, substitutions, and replacements can be made without departing from the spirit and scope of the present disclosure as defined by the appended claims. Besides, the scope of the present application is not limited to specific examples of processes, machines, manufactures, material components, means, methods, and procedures described in the specification. Those skilled in the art can understand from the disclosure of the present application that existing or future developed processes, machinery, manufacturing, and materials that have the same functions or achieve substantially the same results as the corresponding embodiments described herein can be used according to this disclosure. Accordingly, such process, machine, manufacture, material composition, means, method, or step fall within the protection scope of the present application.

Claims

1. A biochemical test chip, comprising: an electrode unit, including a working electrode and a counter electrode; a reactive layer, electrically connected to the working electrode; a protective layer, electrically connected to the working electrode or the counter electrode, wherein the protective layer is configured to oxidate the electrode unit after the electrode unit receives an electron or reduce the electrode unit after the electrode unit loses an electron, wherein a standard reduction potential of the protective layer is higher than a standard reduction potential of the reactive layer; a first insulating septum, located on the electrode unit and including a first opening; and a second insulating septum, located on the first insulating septum and

- including a second opening, wherein the protective layer is vertically aligned with the first opening and the second opening, the first insulating septum is between the protective layer and the second insulating septum, a portion of the first insulating septum and a portion of the second insulating septum form a capillary structure located at a front side of the first insulating septum and the second insulating septum, and the capillary structure extends from outer peripheral edges of the first insulating septum and the second insulating septum, and the first and second openings are positioned over the protective layer and are separated from and fluid incommunicable with the capillary structure.
2. The biochemical test chip of claim 1, wherein a potential difference (E.sub.cell.sup.0) between the protective layer and reactive layer is greater than 0.
 3. The biochemical test chip of claim 1, wherein the protective layer is a cathode, and the electrode unit is an anode.
 4. The biochemical test chip of claim 1, wherein the first opening is between the protective layer and the second opening.
 5. The biochemical test chip of claim 1, wherein the capillary structure at least partially exposes the reactive layer.
 6. The biochemical test chip of claim 1, wherein the electrode unit comprises a branch, wherein the branch is configured to provide the protective layer a conductive platform.
 7. The biochemical test chip of claim 1, wherein the electrode unit and the protective layer are substantially located at a same level.
 8. The biochemical test chip of claim 1, wherein at least a portion of the electrode unit is made of an active material and is in contact with air.
 9. The biochemical test chip of claim 8, wherein the protective layer is configured to protect the portion of the electrode unit that is made of the active material or the whole electrode unit.
 10. A biochemical test chip, comprising: a reactive layer and an electrode unit, wherein the reactive layer is electrically connected to the electrode unit; a protective layer, electrically connected to the electrode unit, wherein the protective layer is electrically connected to the reactive layer via the electrode unit, and the protective layer is configured to oxidate the reactive layer after the reactive layer receives an electron or reduce the reactive layer after the reactive layer loses an electron, wherein there is a potential difference (E.sub.cell.sup.0) between the protective layer and the reactive layer; a first insulating septum, located on the electrode unit and including a first opening; and a second insulating septum, located on the first insulating septum and including a second opening, wherein the protective layer is vertically aligned with the first opening and the second opening, the first insulating septum is between the protective layer and the second insulating septum, a portion of the first insulating septum and a portion of the second insulating septum form a capillary structure located at a front side of the first insulating septum and the second insulating septum, and the capillary structure extends from outer peripheral edges of the first insulating septum and the second insulating septum, and the first and second openings are positioned over the protective layer and are separated from and fluid incommunicable with the capillary structure.
 11. The biochemical test chip of claim 10, wherein the potential difference (E.sub.cell.sup.0) is greater than 0.
 12. The biochemical test chip of claim 10, wherein the protective layer is an anode, and the reactive layer is a cathode.
 13. The biochemical test chip of claim 10, wherein the protective layer is a cathode, and the reactive layer is an anode.
 14. The biochemical test chip of claim 10, wherein an oxidation or reduction reaction current level of the reactive layer is greater than an oxidation or reduction reaction current level of the protective layer.
 15. The biochemical test chip of claim 10, wherein an oxidation or reduction reaction current level of the reactive layer is greater than or equal to 10-fold an oxidation or reduction reaction current level of the protective layer.
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